

Question_7:

Code:

```
1  /*
2  Question No: 7
3  Byteland has N cities connected by N-1 bidirectional roads,
4  so the cities form a tree.
5  The degree of a city is the number of roads connected to it.
6  Two cities are similar when their degrees are equal.
7  A trip is the sequence of cities visited on the shortest path
8  between two cities. Two trips are similar if the corresponding
9  cities in both trips have the same degrees.
10 Find the maximum number of different trips.
11 */
12
13 #include <bits/stdc++.h>
14 using namespace std;
15
16 struct state {
17     int x;
18 };
19
20 const int MAXL=200005;
21 state st[MAXL];
22
23 vector<int> g[MAXL];
24 int d[MAXL];
25 set<vector<int>> s;
26
27 void dfs(int u,int p,vector<int>& v) {
28     v.push_back(d[u]);
29     s.insert(v);
30
31     for(int x:g[u])
32         if(x!=p)
33             dfs(x,u,v);
34
35     v.pop_back();
36 }
37
```

```
38  int main() {
39      int n;
40      cin>>n;
41
42      for(int i=1;i<n;i++) {
43          int u,v;
44          cin>>u>>v;
45          g[u].push_back(v);
46          g[v].push_back(u);
47          d[u]++;
48          d[v]++;
49      }
50
51      for(int i=1;i<=n;i++) {
52          vector<int> v;
53          dfs(i,0,v);
54      }
55
56      cout<<s.size();
57  }
```

Output:

Test Case_1:

```
3
2 1
3 1
5
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
4
2 1
3 1
2 4
8
PS C:\Users\Prabin Yadav\
```